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# A comparison of $PM_{10}$ monitors at a Kerbside site in the northeast of England

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**Abstract**

There is a need for a consistent measurement technique for both $PM_{10}$ and $PM_{2.5}$ that is capable of providing real-time data suitable for determining the effects of particulate pollution on human health. Rupprecht and Patashnik have developed a TEOM® monitor configuration that increases collection efficiency for the semi-volatile mass fraction, when present. By operating at a lower setpoint temperature the system offers a real-time monitor that removes particle bound moisture and promises to improve comparability with the European Union (EU) reference gravimetric method. Trials with the device, a conventionally operated TEOM and a Partisol® gravimetric monitor have shown that in the northeast of England the loss of organics and nitrates may not be the major cause of the observed differences between the monitors. Instead the data presented in this study indicate that it is the retention of particle bound water by the EU reference method that may be causing the observed differences.

The presence and amounts of moisture associated with particles depends on the chemical composition and size range of the particles as well as the ambient relative humidity. As both of these factors vary spatially and temporally it is problematic to apply scaling factors to make data collected by the TEOM comparable to data collected by the EU reference method. In addition, whether particle bound moisture, some of which may be absorbed after sampling should be included in air quality standards needs further investigation.
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## 1. Introduction

Within the European Union (EU) the specified reference method for monitoring $PM_{10}$, EN12341, is a gravimetric technique which requires that the $PM_{10}$ mass fraction is collected upon a filter. Filters are normally exposed for 24 h, although this can be extended if the concentration of particulate matter is thought to be low. Preparation of filters for exposure must be carried out under strict, specified conditions. Prior to, and after exposure, equilibration must take place at $20 \pm 1^{\circ}C$ and a relative humidity of $50 \pm 5\%$ for 24 h (unless further equilibration is indicated). Filters have to be weighed using a balance accurate to within 10 μg (BSI, 1998). Under the terms of the first daughter directive on air quality, 1999/30/EC, EU member states must monitor concentrations of atmospheric $PM_{10}$ using methods that demonstrate equivalence to the reference method. Member states may use alternative techniques providing they can illustrate a consistent relationship to the reference method. Any results obtained must be adjusted by a relevant factor to show equivalence with the results that would have been obtained, should the reference method have been used.

The requirement for filter equilibration and the nature of the operation of gravimetric monitors means that reference method results are often not obtained for a

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period of up to 2 weeks. However, in terms of the determination of health effects and the issue of public information, there is a need for real-time data. In order to meet this requirement many countries are adopting the tapered element oscillating microbalance (TEOM Monitor), which has been in use world-wide since 1988, to give an automatic, near real time measurement of particulate matter (Meyer et al., 2000).

The TEOM method is based on the simple physical law that the frequency of mechanical oscillation of an element is directly proportional to the mass of an element. The TEOM method uses a standard regulatory $PM_{10}$ inlet, operating at $16.7\,l\,min^{-1}$, to select the particles. These then pass through an isokinetic flow splitter from which $3\,l\,min^{-1}$ passes through the 16 mm diameter filter cartridge (13.5 mm active area sampling diameter) connected to the top of a narrow oscillating hollow tapered glass tube. The collection of particles by the filter at the free end of the tapered glass tube will alter its effective mass, which in turn will change its resonant frequency. As more particles deposit on the filter the tube's natural frequency of oscillation decreases (QUARG, 1996). A microprocessor converts the oscillation frequency to mass and then to mass concentrations, which are updated every 2 s. The inlet and the sensing system of the TEOM are operated at a constant temperature of 50°C in order to minimise the adsorption/desorption effects of atmospheric moisture on the microbalance filter (Mukerjee et al., 1999).

Concerns have arisen about the TEOM monitor due to the default operating temperature of 50°C. It is believed that in addition to the removal of water vapour, organic compounds and a proportion of the sulphate and (more likely) nitrate may also be lost. This has resulted in the TEOM monitor sometimes appearing to give lower concentrations than the reference method in Northern Europe (Muir, 2000).

A number of studies have presented data concerning the loss of semi-volatile species on the filter of the TEOM relative to the manual gravimetric method (Allen et al., 1997; Ayers et al., 1999; Meyer et al., 1992). APEG (1999) concluded that in the UK at concentrations around $50\,\mu g\,m^{-3}$ the TEOM tends to under-read compared with a gravimetric sampler by between 15% and 30%. In areas where semi-volatile components constitute a high fraction of $PM_{10}$ it would be desirable to maintain an operating temperature as low as possible, or as close to the reference method equilibration temperature, to minimise the loss of these semi-volatile fractions.

In order to provide a real-time monitor that compares more readily with the EU reference method, Rupprecht and Patashnick have developed a system that enables the sample air stream flowing into a TEOM monitor to be operated at a lower temperature. The sample equilibration system (SES) continuously conditions the flows of the TEOM monitor to remove moisture, allowing the instrument to be operated year-round at a temperature closer to that of the gravimetric reference method (Meyer et al., 2000). A reduction in the operating temperature of the TEOM monitor should allow the greater retention of semi-volatile material.

In order to remove moisture the SES incorporates a Nafion® dryer, which continuously conditions both the sample and bypass flows of the TEOM monitor. The SES enables filter-based mass measurements by the TEOM monitor under consistent, dry conditions (Meyer et al., 2000) and at a reduced temperature. In a validation study using a monitor fitted with a Nafion drier Eatough et al. (2003) have demonstrated that fine particles are not lost during the passage of air through the drier system.

Initial trials with a TEOM monitor fitted with an SES, to monitor $PM_{2.5}$, were carried out in Albany, New York, USA (Meyer et al., 2000). In this study it was found that the TEOM monitor plus SES, operated at 30°C, consistently recorded a higher concentration of particulate matter than the TEOM monitor operating at 50°C. The SES TEOM monitor also showed a better relationship to a gravimetric monitor than the conventional TEOM monitor configuration.

One of the first sites in the EU to monitor particulate matter utilising an SES modified TEOM was Trimdon Street, Sunderland, northeast England. At a kerbside monitoring station, $PM_{10}$ has been monitored using three machines: a TEOM Series1400a monitor operating at 50°C, a Partisol—Plus model 2025 air sampler, and a TEOM series 1400a PM monitor fitted with an SES, operating at 30°C. This paper reports the findings of a trial conducted between November 2000 and August 2001.

## 2. Methods

### 2.1. Sites

All three monitors were sited at the City of Sunderland kerside monitoring unit (sited 1 m from the kerb) in the northeast of England. The site is situated adjacent to a busy road junction with traffic flows of 2000 vehicles $h^{-1}$. Traffic flows are consistently high throughout the day. Fig. 1 illustrates the location of Sunderland within the northeast of England and the location of the sampling site within the City.

### 2.2. Monitors

Monitoring for $PM_{10}$ was carried out using a Partisol Plus 2025 air sampler, a tapered element oscillating

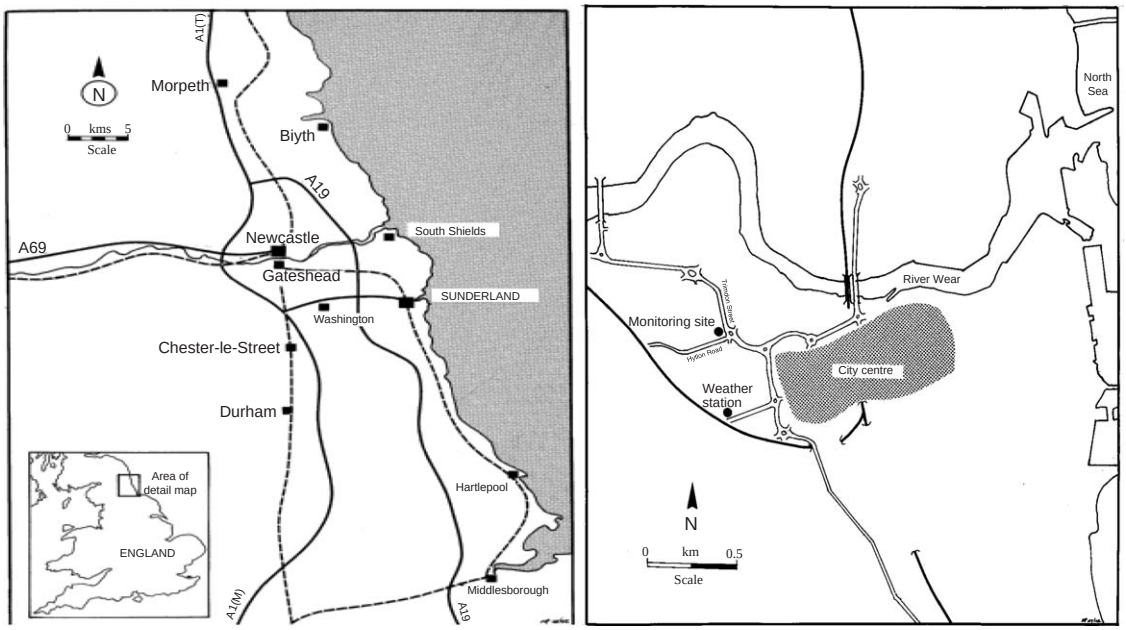



Fig. 1. Location of Sunderland in the northeast of England and the site within the city of Sunderland.

microbalance TEOM series 1400 operating at 50°C and a TEOM monitor plus SES drier operating at 30°C, all manufactured by Rupprecht & Pataschnik Co. The addition of a drier to the TEOM device has been described elsewhere (Meyer et al., 2000). Monitoring using the co-located devices commenced at the site in November 2000. Total nitrogen oxides, nitric oxide and nitrogen dioxide are also monitored at the site by Sunderland city council.

Partisol filters were equilibrated pre- and post-sampling according to the requirements of EN12341 and weighed using a balance accurate to 0.001 mg, with replicate weighings of each filter being made. Initial investigations of the method had shown that there was no further weight loss from filters after the recommended equilibration period. Filters were exposed in the monitor for a period of 24 h.

Flow rates through both TEOM monitors are $16.7\,l\,min^{-1}$ with $3\,l\,min^{-1}$ passing through the sampling head to a 13.5 mm filter. The Partisol has a flow rate of $16.7\,l\,min^{-1}$ passing through a filter of 47 mm diameter (active area of 42 mm), thus the filter face velocities of the air passing through the filters are comparable. Particulate matter was collected onto PTFE-coated glass fibre filters (Pall Corporation).

Meteorological data for the sampling period was obtained from the University meteorological station in Sunderland.

## 3. Results and discussion

### 3.1. Analysis of $PM_{10}$ data

The $PM_{10}$ data collected by the three monitors appears in Figs. 2–4 and the correlation coefficients in Table 1. Good correlation is displayed between the TEOM and the TEOM plus SES all year round. However, the correlation between these monitors and the Partisol sampler appears to be seasonal, with the strongest correlation occurring during the summer months. Figs. 2–4 display the time series for each sampler (where the TEOM and TEOM plus SES are averaged over 24 h periods to match the sampling period of the Partisol). During periods of elevated $PM_{10}$ in the winter and spring, the Partisol sampler records higher mass than the TEOM and the TEOM plus SES. However, during the summer all three instruments tend to record elevated $PM_{10}$ concentrations.

The wintertime recorded differences between the Partisol and TEOM monitors occurred during episodes of elevated $PM_{10}$, with the gravimetric monitor recording a greater increase than the TEOM monitor. The wintertime discrepancy in $PM_{10}$ concentrations as measured by the Partisol and TEOM has been observed in previous studies such as that by Green et al. (2000).

This seasonal difference in the relationship is shown in the strength of correlations (Table 1). In November the correlation coefficient for the relationship between the

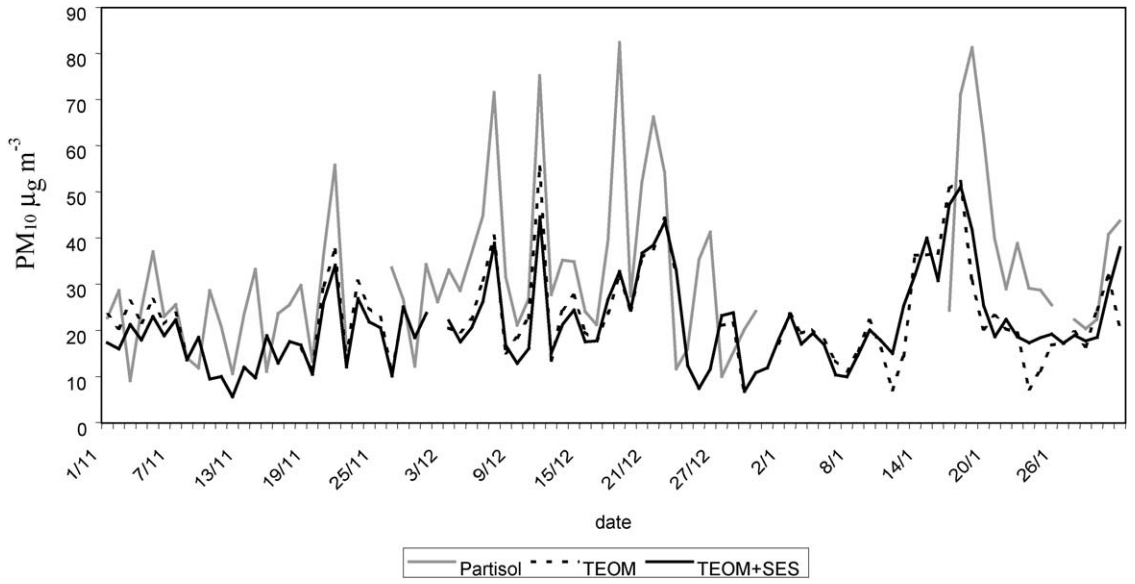



Fig. 2. $PM_{10}$ ($\mu g\,m^{-3}$) results for November 2000–January 2001.

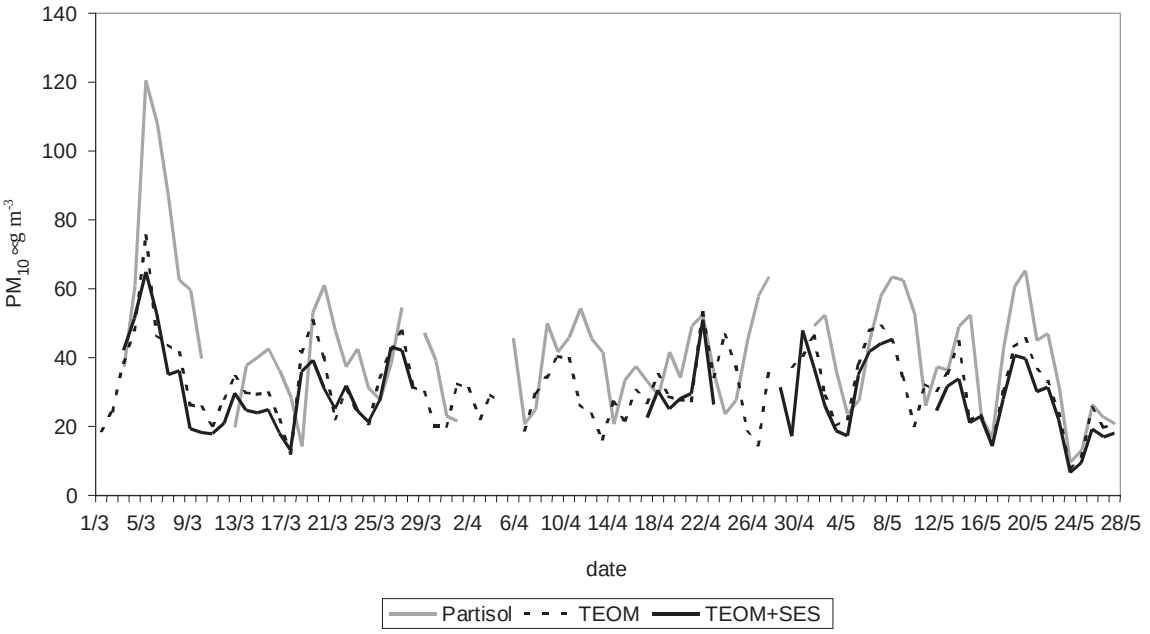



Fig. 3. $PM_{10}$ ($\mu g\,m^{-3}$) results for March 2001–May 2001.

Partisol monitor and the TEOM monitor is 0.59 whereas in August it is 0.89. Throughout the time period the results obtained for the TEOM and the TEOM plus SES show a strong relationship.

Recorded differences between the TEOM and gravimetric monitors are well documented. They include the work of Salter and Parsons (1999), who used a TEOM 1400a series monitor operating at $31\,min^{-1}$ airflow and a Partisol Model 2000 air sampler operating at $16.7\,l\,min^{-1}$. In this study the TEOM was operated at 50°C and results were collected for $PM_{10}$ over a period of 100 days from April to October 1997. The results obtained showed that although the TEOM and Partisol results correlate well at low values of $PM_{10}$ as particulate

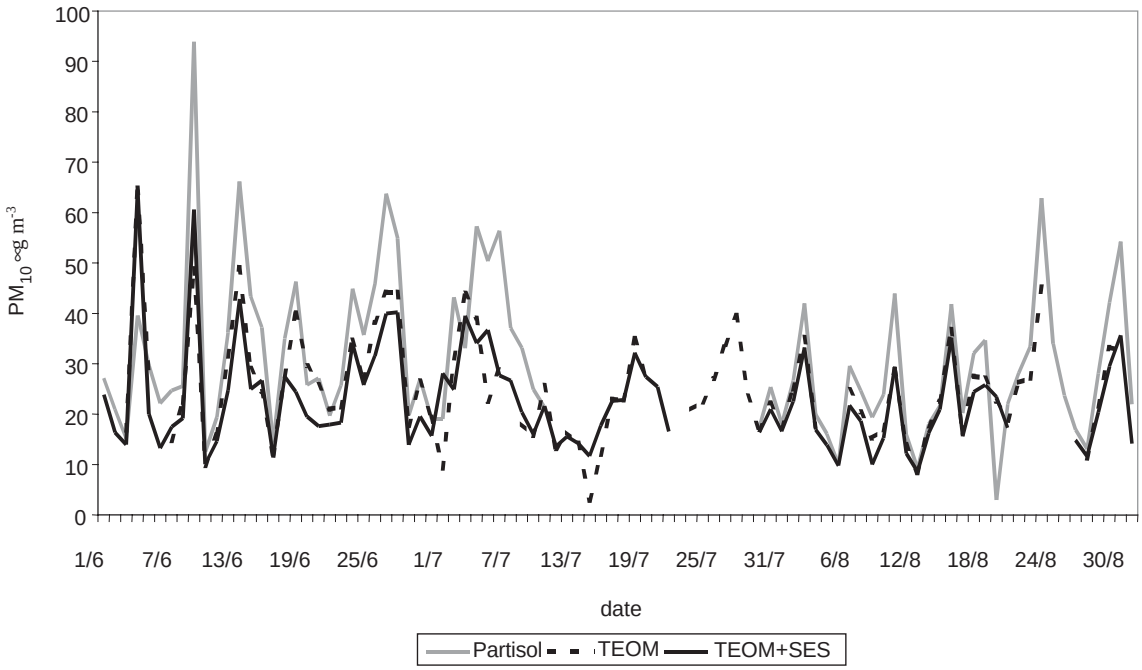



Fig. 4. $PM_{10}$ ($\mu g\,m^{-3}$) results for June 2001–August 2001.

Table 1
Correlation coefficients for the different monitors

| | | TEOM | SES |
|---|---|---|---|
| Full data set | Partisol | 0.675[a] | 0.743[a] |
| | TEOM | — | 0.867[a] |
| November | Partisol | 0.589[a] | 0.502[a] |
| | TEOM | — | 0.879[a] |
| December | Partisol | 0.746[a] | 0.680[a] |
| | TEOM | — | 0.952[a] |
| January | Partisol | 0.403[a] | 0.608[a] |
| | TEOM | — | 0.867[a] |
| March | Partisol | 0.668[a] | 0.664[a] |
| | TEOM | — | 0.931[a] |
| April | Partisol | — | 0.650[a] |
| | TEOM | — | 0.936[a] |
| May | Partisol | 0.737[a] | 0.845[a] |
| | TEOM | — | 0.918[a] |
| June | Partisol | 0.794[a] | 0.841[a] |
| | TEOM | — | 0.932[a] |
| July | Partisol | — | — |
| | TEOM | — | — |
| August | Partisol | 0.887[a] | 0.854[a] |
| | TEOM | — | 0.975[a] |

[a] Significant correlation at $p \leqslant 0.05$.

concentrations increased the Partisol was found to record a higher concentration than the TEOM. A study by Cyrys et al. (2001) evaluated the difference between a TEOM 1400a series monitor operated at 50°C and air flow of $31\,min^{-1}$ with a Harvard–Marple impactor, using Teflon membrane filters (PTFE). Data were collected for the period March–April 1999 and the TEOM consistently recorded lower values than the Harvard Impactor.

Both of the above studies collected data over a relatively short time period even though the composition of particulate matter is reported to vary on a seasonal basis (Harrison et al., 1997). The differing composition of sampled $PM_{10}$ may lead to both spatial and temporal differences in the recorded discrepancies between monitors. The results collected in this study illustrate the seasonal differences and highlight the necessity for monitoring over an extended time scale.

In terms of the compositional differences between $PM_{10}$, as sampled by gravimetric monitors and the TEOM monitor operated at 50°C, a number of studies, including those by Green et al. (2001), Chung et al. (2001) and Allen et al. (1997) suggest that the elevated temperature of the TEOM monitor causes the loss of semi-volatile material including sulphate and nitrate. The results from the present study would suggest that this is not the case at the Sunderland site. The TEOM monitor plus SES drier operated at 30°C, records a comparable value to the TEOM monitor operated at 50°C, and for some periods of the study this was lower.

These results contrast to those reported by Meyer et al. (2000) in an initial trial with the SES carried out at Albany, New York, USA during the summer. During the Albany study temperatures were high, average daytime temperature was 30°C dropping to 10–15°C

at night. The recorded relative humidity was high at night reaching 95–100% but lower during the daytime dropping to 40–50%. The results obtained by Meyer et al. showed an increased quantity of $PM_{2.5}$ monitored by the TEOM plus SES monitor as compared to the conventional TEOM monitor. The results obtained by the TEOM plus SES being comparable to those obtained by a gravimetric monitor.

At the Sunderland site, weather conditions are different to those in Albany, during the winter temperatures range from nighttime lows of –5–8°C and daytime temperatures of 5–12°C. Relative humidity is high throughout the day and night (80–100%). In the summer, temperatures are found to range from nighttime lows of 6–20°C, to daytime temperatures of 12–25°C. Relative humidity in the summer is lower ranging from 55–85%. At this site both TEOM monitors correlate well with the gravimetric monitor during the summer but not during the winter. The contrasting results indicate that the difference between $PM_{10}$ monitors will be dependent upon a range of factors, including meteorology. The results from one study should therefore not be applied to another area of contrasting meteorology and particulate composition.

In the winter at the Sunderland site, the TEOM plus SES and the TEOM operating at 50°C show a good correlation, suggesting that the loss of semi-volatile materials may not be occurring at this site.

In order to obtain an accurate measure of atmospheric particulate matter, the retention of semi-volatile material will be of importance if the composition of the atmospheric aerosol comprises materials that will be lost at 50°C, and under certain circumstances this may comprise materials other than nitrate. Moisture could also be considered as a semi-volatile material lost by the 50°C TEOM monitor. The default operating temperature of the TEOM monitor is chosen to minimise the amounts of particle bound water monitored whereas gravimetric monitors rely on filter equilibration to remove moisture.

A study carried out in Canada (Mignacca and Stubbs, 1999) showed that a reduction in the operating temperature of the TEOM from 50°C to 30°C resulted in a 22% increase in measured $PM_{10}$. However, it is unclear whether this increase is due to the retention of particle bound moisture or semi-volatile nitrate and organics. Without a full knowledge of the composition of the atmospheric aerosol it is not possible to determine the nature of the differences recorded in terms of the composition of the material lost. In some areas the material lost could be organics or nitrate, in other areas it may be particle bound moisture.

Atmospheric aerosol is a complex mixture of inorganic and organic components where organic species can represent up to 50% of the aerosol mass depending on location (Cruz and Pandis, 2000). Whilst the water uptake of the inorganic fraction has been widely investigated and is well understood the knowledge gained concerning the organic fraction has been limited.

In terms of the inorganic fraction at a low relative humidity aerosol particles comprising inorganic salts are solid (Seinfeld and Pandis, 1998). However, as the relative humidity increases a threshold value will be reached above which water will be rapidly absorbed by the particle. The relative humidity at which this occurs is characteristic for the aerosol composition and is termed the deliquescence relative humidity. Values for typical components of atmospheric aerosols are: NaCl—75.3%, $NH_4NO_3$—61.8%, $(NH_4)_2SO_4$—79.9% (Seinfeld and Pandis, 1998). These are all values of relative humidity exceeded in Sunderland routinely in the winter but also during the spring and summer. Seinfeld and Pandis (1998) note that not all aerosol species exhibit deliquescence behaviour, some such as $H_2SO_4$ are hygroscopic and the water content associated with them changes smoothly as the relative humidity increases or decreases.

Studies with the organic fraction have shown that a significant fraction of the particulate organic carbon in the atmosphere is water soluble organic carbon (Yang et al., 2003) which due to its affinity with water plays a role in aerosol–cloud interactions, wet scavenging and the formation of atmospheric haze. The affinity of the material for water should also perhaps be considered in terms of particle bound moisture and the measurement of $PM_{10}$.

Whilst the effects of moisture on the inorganic and organic fraction has been considered the effects of moisture absorption by internal mixtures of salts and insoluble components, for example soot, is more difficult to predict (Ebert et al., 2002). The authors report that organic components can alter the hygroscopic behaviour of salt particles severely.

The quantity of water associated with $PM_{10}$ will therefore be dependent on its composition and the relative humidity of the atmosphere. This is likely to vary both seasonally and temporally and with local and regional pollution sources. The role of particle bound moisture in mixed aerosols is an area that requires further research not only in the atmosphere but also by particles collected on the filters of particulate monitors.

Whilst the TEOM is operated at 50°C in order to minimise particle bound water, equilibration of Partisol filters according to EN12341(BSI 1998) is carried out to minimise the loss of volatile material which may include particle bound moisture. The results obtained in the present study suggest that any water associated with the $PM_{10}$ collected on the filters may also be retained. The correlation between the 30°C TEOM plus SES drier and the TEOM operated at 50°C and the observed differences to the Partisol, most noticeably in the winter, would suggest that moisture plays a role in the differences between monitors at the Sunderland site.

As noted previously the deliquescent relative humidity for particulate matter commonly found in the atmosphere is of the order of 70% RH. However, although the particles will absorb moisture at a relative humidity of 70% they do not start to release this moisture until a much lower RH is achieved, the hysteresis phenomenon. For $(NH_4)_2SO_4$ this may not occur until the relative humidity is reduced to 30–40% (Seinfeld and Pandis, 1998) hence the equilibration relative humidity chosen for Partisol filters (50%) may not be sufficiently reduced to ensure moisture is removed from the particulate mass. If this is the case the differences between gravimetric monitors and real-time monitors, operated to minimise moisture, will be dependent on the amount of moisture associated with the particulate matter collected on the filter of the gravimetric monitor. This in turn will be determined by the composition of the aerosol and atmospheric relative humidity.

### 3.2. Relationship to meteorological parameters

In order to review the role of particle bound water further, the data have been compared to the meteorological data collected for the same period. Tables 2 and 3 present data for the meteorological conditions at times when the greatest discrepancy ($>20\,\mu g\,m^{-3}$) is found between the monitors and also at times of good agreement ($<5\,\mu g\,m^{-3}$). The meteorological data were averaged for the 24 h period of interest to allow comparison with the $PM_{10}$ data.

Rainfall was recorded on several of the days when high discrepancies between the Partisol and TEOM and TEOM plus SES monitors were observed. In addition high $NO_x$ concentrations have been recorded on all of the dates with the exception of 9/6, suggesting that both pollution levels, including a high proportion of fresh aerosol and atmospheric moisture have been high at the same time. The effect of the timing of a rainfall event on the particle bound moisture content of aerosols collected on filters remains unclear. For example, if rainfall occurs before sampling, the particles may be washed out of the atmosphere thus reducing their concentration. However, if rainfall occurs after the filter sample has been collected by the gravimetric monitor, the movement of a moist air stream over the collected particles could lead to the absorption of moisture on both particles and the filter. Of the remaining dates other than 21/12 the dew point had been reached indicating condensation of atmospheric moisture will be occurring.

The timing of increased atmospheric moisture content and the presence of high particle concentrations may explain the contrasting results obtained in the initial study carried out with the TEOM plus SES (Meyer et al., 2000) and the results obtained in Sunderland. In the initial trials relative humidity was high at night and reduced during the day whereas in Sunderland relative

Table 2
Dates of greatest discrepancy ($>20\,\mu g\,m^{-3}$) between the Partisol and TEOM monitors

| Date | Partisol ($\mu g\,m^{-3}$) | TEOM ($\mu g\,m^{-3}$) | TEOM+SES ($\mu g\,m^{-3}$) | $NO_x$ (ppb) | Min temp (°C) | Max temp (°C) | Dew point (°C) | RH (%) | Rainfall (mm) | Rainfall (h) | Sunshine (h) | Mean wind speed ($m\,s^{-1}$) | Wind direction |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 7/12/00 | 71.5 | 40.9 | 39.3 | 108 | 6.4 | 10.0 | 6 | 95 | 16.0 | 7.2 | 0.6 | 3.8 | SW |
| 11/12/00 | 75.2 | 55.8 | 45.0 | 94 | 6.8 | 12.6 | 9 | 86 | 0 | 0 | 2.4 | 3.8 | SW |
| 18/12/00 | 82.4 | 32.2 | 33.1 | 147 | −0.5 | 5.4 | 0 | 98 | 0 | 0 | 0 | 0.7 | S |
| 21/12/00 | 66.2 | 38.1 | 38.9 | 93 | 6.6 | 7.2 | 6 | 91 | 0 | 0 | 0 | 2.8 | SE |
| 18/1/01 | 81.3 | 31.1 | 42.1 | 131 | 0.7 | 3.2 | 1 | 89 | 4.9 | 5 | 0 | 0.5 | Calm |
| 6/3/01 | 108 | 46.5 | 52.9 | 68 | 0 | 7.4 | 0 | 82 | 3.3 | 2.5 | 3.4 | 4.3 | SE |
| 7/3/01 | 87.2 | 43.7 | 35.3 | 107 | 1.7 | 14.3 | 7 | 100 | 0 | 0 | 5.3 | 2.2 | Calm |
| 9/6/01 | 93.9 | 48.9 | 60.6 | 23 | 7.9 | 15.3 | 8 | 85 | 14.5 | 7.6 | 1.7 | 2.7 | W |

Table 3
Dates of good agreement ($<5\ \mu g\ m^{-3}$) between the Partisol and TEOM monitors

| Date | Partisol ($\mu g\ m^{-3}$) | TEOM ($\mu g\ m^{-3}$) | TEOM + SES ($\mu g\ m^{-3}$) | $NO_x$ (ppb) | Min temp (°C) | Max temp (°C) | Dew point (°C) | RH (%) | Rainfall (mm) | Rainfall (h) | Sunshine (h) | Mean wind speed ($m\ s^{-1}$) | Wind direction |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6/11/00 | 23.0 | 21.4 | 19.3 | 28 | 4.8 | 9.6 | 4 | 96 | 23.3 | 16 | 0 | 6.1 | Calm |
| 16/12/00 | 21.2 | 18.1 | 18.1 | 50 | −1.0 | 4.2 | −4 | 74 | 0 | 0 | 1.2 | 1.9 | W |
| 29/1/01 | 22.4 | 24.9 | 18.9 | | 0.4 | 6.6 | 1 | 91 | 0 | 0 | 7.7 | 1.5 | W |
| 24/4/01 | 52.4 | 53.1 | 51.1 | 89 | 3.3 | 10.3 | 7 | 94 | 0.6 | 1.5 | 7.2 | 2.3 | Calm |
| 9/5/01 | 45.1 | 47.9 | 41.9 | 35 | 4.5 | 12.6 | 7 | 84 | 0 | 0 | 2.3 | 2.3 | NE |
| 31/5/01 | 20.8 | 20.6 | 18.3 | 20 | 9.4 | 17.4 | 8 | 66 | 2.3 | 0.4 | 12.6 | 5.3 | W |
| 3/6/01 | 15.2 | 16.8 | 14.0 | 12 | 6.0 | 14.7 | 3 | 58 | 0 | 0 | 6.8 | 3.8 | N |
| 1/8/01 | 26.7 | 24.9 | 22.5 | 35 | 12.0 | 24.5 | 10 | 64 | 0 | 0 | 12.3 | 1.9 | W |
| 14/8/01 | 21.6 | 22.5 | 20.9 | 29 | 18.8 | 24.7 | 17 | 70 | 0.7 | 0.4 | 10.1 | 3.9 | SW |

humidity is high throughout both the day and night. At the Sunderland site the RH is high when atmospheric particulate pollution is high whilst in the Albany study the relative humidity was rising as particulate levels may have been falling.

Table 3 shows that the meteorological conditions found when there is a good agreement between the monitors have tended to be dry with some sunshine recorded. However, there are occasions when rainfall has been recorded or the dew point reached and there is no discrepancy between the monitors. The variation in composition of atmospheric particulate matter will determine the amounts of moisture adsorbed and this could account for the variability in results obtained.

A thin film of water molecules coats many substances under ambient conditions (Romakkaniemi et al., 2001) and adsorption has been shown to occur at relative humidities below the deliquescent relative humidity. Due to its importance, the hygroscopic properties of aerosol particles has been studied for a long time (Ebert et al., 2002) including the work of Hanel (1976). A number of studies have considered the adsorption of moisture by atmospheric particulate matter including that by Hameri et al. (2000), who have measured the hygroscopic growth of ultrafine aerosol particles. However, in the gravimetric measurement of atmospheric $PM_{10}$ and $PM_{2.5}$ it is assumed that once a particle is deposited on a filter it remains inert even in the presence of an air stream of high relative humidity containing a range of gaseous pollutants. The work of Hameri et al. (2000) and Romakkaniemi et al. (2001) would suggest that this is unlikely to be the case and that collected particles will continue to adsorb moisture at relative humidities at or above the deliquescent relative humidity for the sampled particles. In Sunderland, particularly during the winter, the recorded ambient relative humidity is above the deliquescent relative humidity for particulate matter commonly sampled by gravimetric monitors.

Although the gain of moisture by atmospheric particulate matter associated with filters has been illustrated by a number of researchers including the work of Jarrett et al. (2001) its subsequent removal or retention during filter conditioning has yet to be determined. When the relative humidity decreases the deliquescent RH will be reached, however, the particle does not lose all of its associated moisture at this or a decreased relative humidity (Seinfeld and Pandis, 1998). In order to regain a solid state, salt nuclei need to be formed and salt crystals must grow around them. In the atmosphere this occurs at a relative humidity significantly lower than the deliquescent relative humidity. For salts such as ammonium nitrate with a deliquescent relative humidity of 61.8% this may well be below the relative humidity of 50% chosen for filter conditioning.

## 4. Conclusion

Particulate matter both $PM_{10}$ and $PM_{2.5}$ will comprise a range of chemical species plus variable amounts of particle bound water, where the amount of moisture will be dependent on the composition of the atmospheric aerosol and the relative humidity of the atmosphere. This will vary both temporally and spatially.

The default operating temperature chosen for the TEOM of 50°C, minimises the amounts of particle bound moisture monitored. This has led to criticisms that the TEOM under-records atmospheric particulate matter due to the loss of semi-volatile materials such as organics and nitrates. However, particle bound moisture could be classified as a semi-volatile constituent of atmospheric aerosols which is lost by the TEOM but retained by gravimetric monitors. The results of the monitoring campaign carried out in Sunderland, using a TEOM fitted with a drier and operated at a lower temperature, would suggest that at this site the gravimetric monitor is retaining a greater proportion of atmospheric moisture than the TEOM monitor.

At this site it would therefore not be possible to apply a factor to adjust the TEOM results to make them comparable with the EU reference method, in that the recorded differences are not consistent but governed by aerosol composition and atmospheric relative humidity. In addition application of a factor would unnecessarily penalise local industry in that the increased amounts of particulate matter recorded by the gravimetric monitor could be due to atmospheric moisture and not an increased anthropogenic input.

In any review of monitors and air quality standards the role of particle bound moisture needs to be evaluated.

## References


Allen, G., Sioutas, C., Koutrakis, P., Reiss, R., Lurmann, F.W., Roberts, P.T., 1997. Evaluation of the TEOM method for measurement of ambient particulate mass in urban areas. Journal of the Air and Waste Management Association 47, 682–689.

APEG, 1999. Source apportionment of airborne particulate matter in the United Kingdom. Airborne Particulates Expert Group.

Ayers, G.P., Keywood, M.D., Gras, J.L., 1999. TEOM vs. manual gravimetric methods for determination of $PM_{2.5}$ aerosol mass concentrations. Atmospheric Environment 33, 3717–3721.

BSI, 1998. BS EN 12341:1998. Air quality—determination of the $PM_{10}$ fraction of suspended particulate matter—reference method and field test procedure to demonstrate reference equivalence of measurement methods. British Standards Institute.

Chung, A., Chang, D.P.Y., Kleeman, J., Perry, K.D., Cahill, T.A., Dutcher, D., McDougall, E.M., Stroud, K., 2001. Comparison of real-time instruments used to monitor airborne particulate matter. Journal of the Air and Waste Management Association 51, 109–120.

Cruz, C.N., Pandis, S.P., 2000. Deliquescence and hygroscopic growth of mixed inorganic–organic aerosol. Environment Science and Technology 34, 4313–4319.

Cyrys, J., Dietrich, G., Kreyling, W., Tuch, T., Heinrich, J., 2001. $PM_{2.5}$ measurements in ambient aerosol: comparison between Harvard Impactor (HI) and the tapered element oscillating microbalance (TEOM) system. The Science of the Total Environment 278, 191–197.

Eatough, D.J., Long, R.W., Modey, W.K., Eatough, N.L., 2003. Semi-volatile secondary organic aerosol in urban atmospheres: meeting a measurement challenge. Atmospheric Environment 37, 1277–1292.

Ebert, M., Inerle-Hof, M., Weinbruch, S., 2002. Environmental scanning electron microscopy as a new technique to determine the hygroscopic behaviour of individual aerosol particles. Atmospheric Environment 36, 5909–5916.

Green, D., Fuller, G., Barratt, B., 2000. 'Marylebone Road ('Supersite'), Annual Report, 199. http://www.seiph.umds.ac.uk/detr/ss_reports/reportar98.htm.

Green, D., Fuller, G., Barratt, B., 2001. Evaluation of TEOM 'correction factors' for assessing the EU Stage 1 limit values for $PM_{10}$. Atmospheric Environment 35, 2589–2593.

Hameri, K., Vakeva, M., Hansson, H.C., Laaksonen, A., 2000. Hygroscopic growth of ultrafine ammonium sulphate aerosol measured using an ultrafine tandem differential mobility analyzer. Journal of Geophysical Research-Atmospheres 105, 22231–22242.

Hanel, G., 1976. The properties of atmospheric aerosol particles as functions of the relative humidity at thermodynamic equilibrium with the surrounding moist air. Advances in Geophysics 19, 74–183.

Harrison, R.M., Deacon, A.R., Jones, M.R., 1997. Sources and processes affecting concentration of $PM_{10}$ and $PM_{2.5}$ particulate matter in Birmingham (UK). Atmospheric Environment 31 (24), 4103–4117.

Jarrett, R.P., Clark, N.N., Gilbert, M., Ramamurthy, R., 2001. Evaluation and correction of moisture adsorption and desorption from a tapered element oscillating microbalance. Powder Technology 119 (2–3), 215–228.

Meyer, M., Lijek, J., Ono, D., 1992. Continuous $PM_{10}$ measurements in a woodsmoke environment, $PM_{10}$ standards and non-traditional particulate controls. In: Chow, J.C., Ono, D.M. (Eds.), Journal of the Air and Waste Management Association 21 (1), 24–38.

Meyer, M.B., Patashnick, H., Ambs, J.L., Rupprecht, E., 2000. Development of a sample equilibration system for the TEOM continuous PM monitor. Journal of the Air and Waste Management Association 50, 1345–1349.

Mignacca, D., Stubbs, K., 1999. Effects of the equilibration temperature on PM10 concentrations from the TEOM method in the Lower Fraser Valley. Journal of the Air and Waste Management Association 49 (10), 1250–1254.

Muir, D., 2000. The suitability of tapered element oscillating microbalance (TEOMs) for $PM_{10}$ monitoring in Europe. The use of $PM_{10}$ data as measured by TEOM for

compliance with the European Air Quality Standard. Atmospheric Environment 34, 3209–3212.

Mukerjee, S., Shadwick, D.S., Bowser, J.J., Carmichael, L.Y., 1999. Application of the dual fine particle sequential sampler, a tapered element oscillating microbalance and other air monitoring methods to assess transboundary influences of $PM_{2.5}$. Field Analytical Chemistry and Technology 3 (3), 201–217.

QUARG, 1996. Airborne particulate matter in the United Kingdom. Third report of the Quality of Urban Air Review Group. May 1996, Department of the Environment. ISBN 0 0952077132.

Romakkaniemi, S., Hameri, K., Vakeva, M., Laaksonen, A., 2001. Adsorption of water on 8–15 nm NaCl and $(NH_4)_2SO_4$ aerosols measured using an ultrafine tandem differential mobility analyzer. Journal of Physical Chemistry A 105 (35), 8183–8188.

Salter, B., Parsons, L.F., 1999. Field trials of the TEOM and partisol for $PM_{10}$ monitoring in the St Austell china clay area, Cornwall, UK. Atmospheric Environment 33, 2111–2114.

Seinfeld, J.H., Pandis, S.N., 1998. Atmospheric Chemistry and Physics. From Air Pollution to Climate Change. Wiley Interscience, New York.

Yang, H., Qianfeng, L., Jian Zhen, Y., 2003. Comparison of two methods for the determination of water-soluble organic carbon in atmospheric particles. Atmospheric Environment 37, 865–870.